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Acknowledgement

I would like to thank Stephen Hopkins of Shalimar, Florida for the help with this competition rules document. In 2022, IEEE Region 3's SoutheastCon held a robotics competition called "Hurricane Alley 'Cat5 Robots Rebuild the Magic at Duck Gardens'" with a very strong central Florida theme. I edited those rules to encourage our students to tackle a variety of challenges and to incorporate a more Mississippi style. I am so thankful to Stephen for his support of IEEE and Mississippi State University ECE's SECON team.

-Ryan Green

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I. Introduction

Starkville Mississippi has seen some really bad weather. If the flood hadn't been enough, the tornado was just adding insult to injury! This storm has caused significant damage. Particularly, Starkville's duck-based theme park, *Quackerjack's Hollow*, is in a real mess. The owners of *Quackerjack's Hollow* have come to Mississippi State University's Electrical and Computer Engineering Department looking for a solution. They are not only looking for immediate cleanup and rebuilding strategies, but also longer-term strategies to rebuild if such a storm were to hit again.

We need autonomous robots that can help animals (ducks, alligators, raccoons), clean up debris and rebuild this beloved (and totally real) theme park. Student teams are to design and implement these storm cleanup robots to recover *Quackerjack's Hollow*. There are several tasks to do to help restore the Golden Triangle's only duck-inspired theme park. The robot may do them all or as few as they would like to do, but *Quackerjack's Hollow's* executives will purchase the student-designed robot that performs the most tasks and meets most, if not all of, their goals.

II. Playing Field

The main course is constructed of a 4' x 8' sheet of plywood with a boundary of 1" x 4" wooden beams on three sides and two 2" x 4" x 4' wooden beams on one shorter end. There is also another 2' x 2' board on the center of the southern part of the board for the park's ticket booth, where robots enter the park. The base of the board left unpainted, but has areas of color for the

different “attractions” at *Quackerjack’s Hollow*. **Error! Reference source not found.** includes the design of the board. **Error! Reference source not found.** shows the constructed board.

Table 1 shows the board design legend.

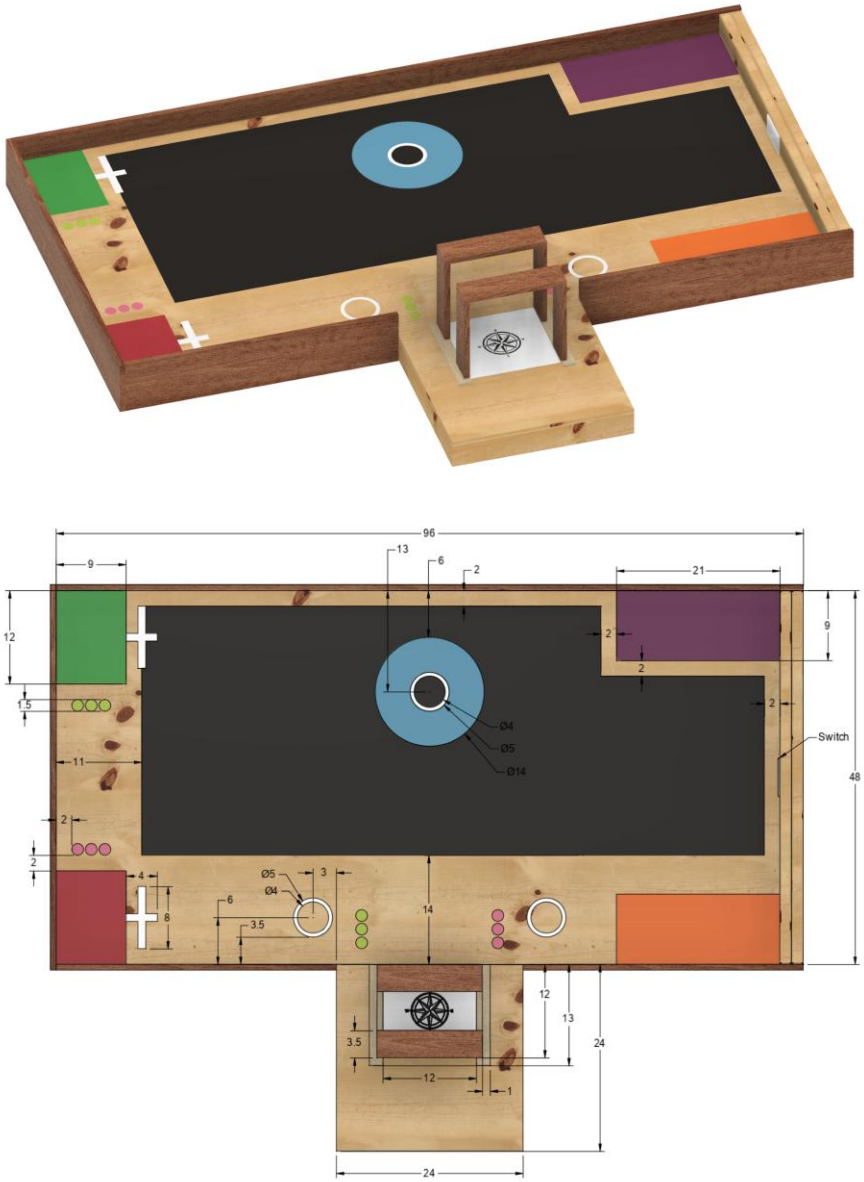


Figure 1. Board Design (Top) Overview without dimensions. (Bottom)

Forthcoming
Figure 2. Constructed Board Photo

Table 1. Board Design Legend

***Quackerjack's Hollow* Board Component**	**Description**
Total Board	4' x 8' Arena (3' 8 1/8" x 7' 10 3/8" interior)
Green Rectangle	Alligator Aquarium Attraction
Red Rectangle	Raccoon Enclosure Attraction
Blue Circle	Duck Pond Attraction
White circle with Black Ring and White Ring	Duck Statue Attraction
Orange Rectangle	Recycling Area South
Purple Rectangle	Recycling Area North
Light Gray Rectangle	Switch Box for Fireworks
Dark Gray Rectangle	Fireworks Display Computer
Black Areas	Asphalt Guest Area
Tan Areas	Dirt Guest Area
Dark Brown Lines	Arena Walls

III. Objective

Teams will build a robot that operates inside *Quackerjack's Hollow*. Unfortunately, while a few broken road lines remain, most roads are washed out. The storm forced the ducks out of the pond and knocked over the three duck statues. The ducks and statue pedestals are scattered throughout the park. The park includes raccoons and alligators in their respective enclosures. The raccoons are hungry, but safe. The alligators are next to the raccoons. They are hungry too, and the raccoons and ducks are starting to look appetizing. The robot will navigate the park feeding raccoons, feeding alligators, moving ducks back to the pond, rebuilding the statues, and placing

the remaining pieces in the recycling bins. In customary theme park tradition, the robots may end their cleanup by activating the park's fireworks display! The robot may do any of these tasks in any order, except starting the fireworks. When the robot initiates the fireworks, the cleanup is over.

IV. Storm Damage

The three types of storm damage we've seen here at *Quackerjack's Hollow* are ducks, statue pedestals, and trash.

4.1 Ducks:

The ducks, seen in Figure 3, have been blown all over the theme park. Here is what we know about them:

- None of the ducks are located inside any attraction or the recycling area, thankfully.
- There is no rhyme or reason as to how they are laying on the ground.
- No ducks are within 2" of an outside wall.
- There are no ducks to the west or east of the theme park entrance. This means the robot can move west towards one end or east towards the other end without running into a duck.
- There are 10 ducks in the theme park: nine are yellow, and one is pink.
- Ducks are found here (link) and here (link).

Figure 3. Yellow Ducks (left) and Pink Duck (right)

4.2 Statue Pedestals:

The statue pedestals, seen in Figure 4, are cylinders. Here is what we know about the statue pedestals:

- None of the pedestals are located inside any attraction or recycling area.
- There is no rhyme or reason as to how they are laying on the ground.
- No pedestals are within 2” of an outside wall.
- There are no pedestals to the west or east of the theme park entrance. This means the robot can move west towards one end or east towards the other end without running into a pedestal.
- There are 7 statue pedestals, which can be stacked to complete 3 total statues.
    - Two statues, each with two levels, will be located to the west and east of the robot’s starting position. Each of these two statues can be completed using:
        - One white pedestal on the first (bottom) level of the statue
        - One green pedestal on the second (top) level of the statue
    - One statue will be located in the duck pond. This statue has three levels and consists of:

        - One white pedestal on the first (bottom) level of the statue
        - One green pedestal on the second level of the statue
        - One red pedestal on the third (top) level of the statue

- The pedestals are made with TPU 1.75 mm diameter filament.
    - The 3D print file is located here (link).
    - The 3D print has a 10% infill.
    - The 3D print filaments are located here (white) (green) (red)

Forthcoming

Figure 4. Pond Statue before the storm. White pedestal (bottom pedestal), Green pedestal (middle pedestal), and Red pedestal (top pedestal)

4.3 Trash:

There will be 5 yellow plastic chips and 5 blue plastic chips designated as “trash pieces”, randomly placed in the park. Here is what we know about the plastic chips:

- None of the trash pieces are located inside any attraction.
- There is no rhyme or reason as to how they are laying on the ground.
- No trash piece is within 2” of an outside wall.

- There are no trash pieces to the west or east of the theme park entrance. This means the robot can move west towards one end or east towards the other end without running into a trash piece.
- There are 10 trash pieces in total.
- The trash pieces will all be the same style of plastic chip (diameter of 1.5”).
- The chips used as trash pieces can be found here (link).

V. *Quackerjack’s Hollow* Cleanup Activities

Quackerjack’s Hollow is a 4’ x 8’ theme park. The entrance to *Quackerjack’s Hollow* is on one 8’ side of the park. The entrance is located halfway along the side. The robot will start just inside the entrance on a 12” x 12” white square painted on the black painted board. The robot must fit within this 12” x 12” square and not be any taller than 12”. The west end of the park includes a raccoon cage and an alligator aquarium (referred to collectively as enclosures) as attractions. The east end of the park includes the recycling areas and the fireworks attraction. Directly in front of the robot’s starting position is the duck pond attraction. There are three duck statue attractions, which are pedestals with a duck on top. One duck statue is in the center of the pond. The other two duck statues are on the west and east sides of the robot’s starting area.

5.1 Duck Statues

There are three duck statues located at *Quackerjack’s Hollow*.

- The robot can choose to rebuild the statues.
- Two statue locations are on either side of the robot starting location.
    - These statues have two pedestals (white for the base and green on top)
    - The statue placed to the west of the robot’s starting area is centered 6” North of the South wall and 33” from the West wall of the park.

    - The statue placed to the east of the robot's starting point is centered 6" North of the South wall and 33" from the East wall of the park.
- The third duck statue location is in the center of the duck pond.
    - This statue has three pedestals (white for the base, green for the middle, and red on top).
    - This statue is 12" from the park's north wall. It is also 28" from either West or East walls.
- Each duck statue location is a circular area with two rings.
    - The inner ring has a radius of 2" from the center and is a black to white transition moving outward from the center of the ring.
    - The outer ring has a radius 2.5" from the center and is a white to black or blue transition moving outward from the center of the ring.
- Each completed statue has a duck on the top of the highest pedestal.
    - Robots placing the pink duck on top of the statue are awarded more points than placing a yellow duck. Placing the pink duck on the tallest (3-pedestal) statue is worth more than placing the pink duck on a shorter statue.
    - Any duck placed on top of a statue is awarded points.
- The areas for each statue are marked with an outer and inner ring. Robots should build statues inside the inner ring without the statues touching anything outside of the inner ring.
    - Fewer points will be awarded for statues that are inside the inner ring but touching the outer ring.
    - Fewer points will apply for statues built entirely within the outer ring
    - No points will apply for statues built anywhere else.

- **The pedestals stacked must be freestanding and not leaning or touching anything else at the end of the round. At the end of the round, if a pedestal falls, these points will not be counted towards the statue score.**

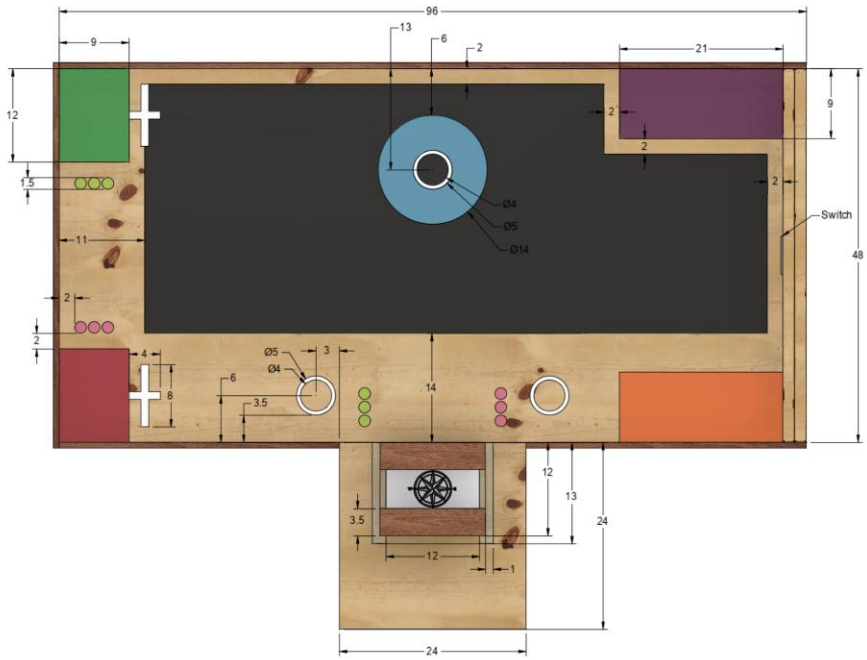
5.2 Raccoon and Alligator Attractions

There are two animal attractions at *Quackerjack's Hollow*.

- The raccoon enclosure is a 9" x 12" rectangular area in the southwest corner of the park. The raccoon enclosure is painted red.
- The alligator aquarium is a 9" x 12" rectangular area in the northwest corner of the park. The alligator aquarium is painted green.
- The 12" sides of both enclosures are oriented in the North-South direction. The western border of the enclosure is aligned with the western wall of the park. The 9" sides of both enclosures are perpendicular to the 12" sides. One 9" wall of each enclosure is aligned with a north or south wall.
- The raccoon enclosure will be in the park's northwest corner while the alligator aquarium will be in the park's southwest corner.
- Just outside the eastern border of each animal enclosure is a white cross painted on the arena floor.
    - The cross is composed of an 8" line and a 4" line. The 8" line runs parallel to the 12" side of the enclosure and is centered 2" outside the enclosure. The 4" line is perpendicular to the 8" line and centered at 6" along the 12" border of the enclosure.
- There will be 6 green food chips for the alligators and 6 red food chips for the raccoons.
    - The teams will be allowed to pre-load up to 6 food chips in the robot (maximum of 3 green and 3 red) next to the starting area near the park entrance.

- If the team chooses to pre-load 6 food chips, before the start of the match, the team will take the 6 chips near the starting position and place them in their bot. The remaining 6 food chips (3 green and 3 red) will be placed near their respective animal enclosures.

If the team chooses to NOT pre-load 6 food chips, all 12 of the food chips (6 green and 6 red) will be placed near the park entrance as seen in



- Figure 1.
- The two lines of 3 red chips will be initially set up as follows:
  - The first line of 3 red chips will be placed 2 inches North of the red enclosure
    - The chips will be placed in a line from east to west, parallel to the 9” side of the enclosure.
    - The chips will be separated by 0.25” from each other along the line.
    - The west-most chip will be 2” from the western border of the park.
  - The second line of 3 red chips will be placed 2 inches to the east of the starting area near the entrance of the park.

        - The chips will be placed in a line from north to south, parallel to the vertical 12” side of the starting area.
        - The chips will be separated by 0.25” from each other along the line.
        - The south-most chip will be 2” from the southern wall of the park.
    - The two lines of green chips will be initially set up as follows:
        - The first line of 3 green chips will be placed 2 inches south of the green enclosure
            - The chips will be placed in a straight line from east to west, parallel to the 9” side of the enclosure.
            - The chips will be separated by 0.25” from each other along the line.
            - The western-most chip will 2” from the western wall of the park.
        - The second line of 3 green chips will be placed 2-inches to the west of the starting area near the entrance of the park.
            - The chips will be placed in a straight line from north to south, parallel to the vertical 12” side of the starting area.
            - The chips will be separated by 0.25” from each other along the line.
            - The southern-most chip will be 2” from the southern wall of the park.
- The robot will be awarded points for putting food chips in the enclosures
    - Robots will be awarded more points for putting the correct food chips in the corresponding color enclosure (red to red, green to green).
    - Any food chips in any enclosure will be awarded some points.
    - The robot does not have to place food chips in either enclosure.
    - The robot may add food chips to one enclosure and not the other.
- The food chips used can be found here (link).

5.3 Duck Pond

We need to clear the park so everyone can get back to the magic of *Quackerjack's Hollow*. Robots will relocate the ducks back to the duck pond.

- The duck pond is the blue circle with the black and white circles in the center. The blue circle has a diameter of 14", is centered horizontally on the board, and is 6" from the northern wall)
- Robots may place 0 to 10 ducks in the pond. If the robot is placing ducks on top of statues, then those ducks cannot be counted as pond dwelling ducks. A duck that falls off a statue before points are finalized will not be counted as a statue duck.

5.4 Recycling Areas

There are two recycling areas at *Quackerjack's Hollow*.

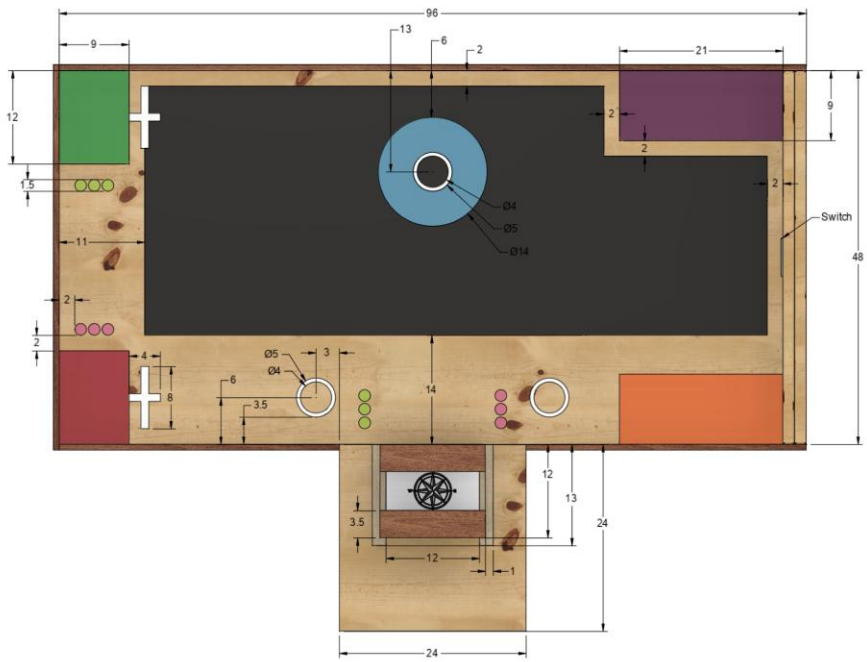
- Each recycling area is a 21" by 9" rectangular area in each corner of the east side of the park. The recycling area on the north side of the board is painted purple. The recycling area.
- Points will be awarded for each trash piece that is moved into either recycling area.
- Fewer points will be awarded for each unused food chip, pedestal, or duck moved into the recycling areas.
- No points will be awarded for materials used to assemble statues in the recycling areas.

5.5 Fireworks

To celebrate a job well done, the owners are allowing us to shoot off fireworks. The fireworks are located on the eastern side of the park. *Quackerjack's Hollow* is set up to start them relatively easily, when you are ready to go.

- A fireworks-activating light switch is mounted horizontally on two 2” x 4” x 4’ wooden beams in a metal electrical box. The switch used can be found here (link). The switch plate used can be found here (link).

The light switch will begin in the off state at the start of each round. In the off state, the switch will be in the upward position with respect to



  - Figure 1, pointing to the northern wall of the park.
  - The light switch may be flipped into the on state during the match. In the on state, the switch will be in the downward position with respect to **Error! Reference source not found.**, pointing to the southern wall of the park.

- The electrical box is a 1-gang, new work/old work standard metal electrical box. The box used is here (link).
  - The switch is connected to a small project board, which will activate a video of fireworks on a computer.
  - When finished with all chosen tasks, the robot may, but is not required to, flip the switch and activate the fireworks attraction.

VI. Vehicle

The robot has defined The robot must be completely autonomous and follow the guidelines below. “Must” guidelines are requirements the robot must have that WILL result in disqualification if not followed. “Should” guidelines are strong suggestions by the competition organizers to prevent potential damages or loss of points. “May” guidelines are allowances by the competition organizers to encourage out of the box thinking. “Cannot” guidelines are requirements that the robot must not have or do that WILL result in disqualification if the guidelines are not followed.

- Must:
    - Must be completely autonomous and must not have any contact with a competitor, spectator, judge, or other person through means including, but not limited to, wireless communication, visual power detection, audio signals, infrared signals, or any other method that is not handled completely autonomously by the platform.
    - Must fit within the 12” x 12” x 12” starting area at the start of each match but may expand beyond the initial size to any size after the match has started.
    - Must be wholly contained within the playing surface at all times of the match but may extend a maximum of 1” beyond the outside edge of the arena walls and a maximum of 0.5” down the backside of the arena walls.
    - Must start the competition run with a single switch or button labeled “Start”.
        - While multiple switches can exist on the robot for powering up and controlling the various subsystems on the robot, there must be a single start switch that is labeled and clearly visible.
    - Must leave the starting area before the first 10 seconds in the round.
        - Leaving the starting area includes expanding beyond the starting area.
        - If the robot has not moved beyond the starting area after the first 10 seconds. the robot forfeits the round.

- Must have an emergency cut-off switch to allow the team to disable the robot if necessary.
    - This emergency cut-off switch must be easily reachable to allow the robot team to disable the robot if necessary.
    - If the robot splits into multiple independent robots, the single emergency cut-off switch should stop the motion of all the robots.

- Should
    - Should automatically stop their work at the 3-minute mark to avoid teams having to manually reach over the top of the robot and accidentally interfere with statues, pedestals, trash pieces, or ducks.
        - All scoring will occur after the team has retrieved their robot from the playing area.

- May:
    - May split apart into multiple independent robots after the match has started.
    - May expand beyond the initial size to any size after the match has started.
    - May weigh what the team can feasibly handle. There is no weight restriction on robots.
    - May use either a toggle switch or push button for the start switch.
    - May display a flag on the robot representing the team name or logo.
        - The flag must fit within the initial size constraints of the robot.
        - The flag can be static or raised automatically at any time once the match begins.
        - The flag can be no bigger than 2” x 4”.
        - The flag cannot contain any objectionable material.

- Cannot:

- Cannot fly or be airborne at any point in the competition. The robot must always remain in full contact with the arena floor during the match. Aerial or flying robots are not allowed.
- Cannot use wireless communication. Any use of wireless communication on the robot is immediate grounds for disqualification. No remote controls are allowed. The team must be ready to show that all wireless communications are disabled before competition.
- Cannot contain any explosive, pyrotechnic, toxic, or corrosive materials. Flammable liquids or gases are also prohibited.
    - Compressed gas is allowed on the robot as long as the pressure is limited to no more than 30 pounds per square inch at any time. Gases other than air are permitted as long as they do not pose a safety threat if accidentally released. All gasses other than plain air must have approval before being incorporated into a design.
- Cannot present any danger to the judges, spectators, competitors, playing arena, itself, or area surrounding the arena.

NOTE: If at any time the judges deem the robot is disqualified via violating the rules stated above, the judges may terminate the round immediately and the team will be required to activate the emergency cut-off switch.

NOTE: The judge will have the discretion of whether any points are awarded for that match and if the robot is allowed to complete any remaining matches.

VII. Rules of Play

The competition will comprise of 3 rounds of 3 minutes of play. The competition will follow the flow enumerated below including sequestration, match pre-staging, setting up the match, conducting the match, scoring, and returning to sequestration.

7.1 Sequestration

- At the start of each round of matches, the judges will require all robots be sequestered in a special staging area.
    - Once in the staging area, the robots must remain turned off, disconnected from other devices, and cannot be accessed by students until they are called for their match to begin.
    - The sequestration judges will assess the robots for compliance with the rules of the competition.
        - If the sequestration judges have assessed the robots to meet the rules of the competition, the robot will remain in sequestration until the team is called for competition play.
        - If the sequestration judges have assessed the robot to not meet the rules of the competition, the robot will forfeit this round and will be asked make necessary changes before they can compete in the next round of matches.
            - Forfeiture means 0 points will be entered for that round for that team.
    - The robots cannot be charging during the sequestration period.

7.2 Match Pre-staging

- For each match, judges and organizers will call the names of the teams to run that match. Once called, teams will have two minutes to retrieve their robot from the sequestration

area and move it to a pre-staging area near the arena. In this area, the team can perform final checks to the robot, make any last-minute adjustments, or swap in a new set of batteries (but not power the robot on). Judges will then call the team from the pre-staging area to the match area when the board is ready to be set up.

7.3 Setting up for a Match

The judges will call teams to witness and inspect the placement of debris (trash pieces, ducks, and pedestals). Judges will place the following into a bucket:

- 9 yellow ducks,
- 1 pink duck,
- 1 red pedestal,
- 3 green pedestals,
- 3 white pedestals,
- 5 yellow trash pieces, and
- 5 blue trash pieces

The bucket will then be emptied by the judges onto the board. All debris that extends outside of the Asphalt Guest Area (Black) will be placed back into the bucket by the judges and emptied again until all debris is placed on the board and none of the debris extends beyond this area. Extending beyond this area means if any item, in a resting state, overlaps black and any other color. As a result, the Asphalt Guest Area does not contain the white cross next to the alligator enclosure (green), the pond, or statue in the pond.

After the debris is administered, the following will happen in no particular order:

- 3 red chips will be placed 2” north of the northern-most edge of the raccoon enclosure and 2” from the western wall of the park. The chips will be placed in a horizontal line with a ¼” separation.

- 3 red chips will be placed 2” to the east of the eastern edge of the starting area and 2” from the southern wall of the park. The chips will be placed in a vertical line with a ¼” separation.
- 3 green chips will be placed 2” south of the southern edge of the alligator area and 2” from the western wall of the park. The chips will be placed in a horizontal line with a ¼” separation.
- 3 green chips will be placed 2” to the west of the western edge of the starting area and 2” from the southern wall of the park. The chips will be placed in a vertical line with a ¼” separation.

Once the debris and food chips are administered on the board, the team will either

(1) Acknowledge that the board is ready or

(2) Suggest items to readjust.

Debris cannot be adjusted for orientation. Debris can be adjusted only if its placement results in the debris extending beyond the Asphalt Guest Area. If the team and judges cannot come to an agreement within 2 minutes, both the teams and judges will exit the arena while an event organizer makes any adjustments needed.

NOTE: When either (1) the team and the judge have agreed on the board configuration or (2) the event organizer has adjusted pieces on the board, no further adjustments can be made to the board configuration.

NOTE: The event organizer may ask for help from a event volunteer not participating in this round if needed.

After the board is set up, teams will have two minutes to retrieve their robot and set it up for the match. At this point, they will:

- move their robot into the starting area.
- set the robot into a standby-mode. The robot must not start running the course.

- let the judges know if they intend to pre-load the 6 food chips closest to the park entrance into their bot.

There is no limit on the number of team members near the arena during the setup as long as they can do so within this setup time.

7.4 Conducting the Match

- Once the team is ready or the two minute time limit has passed, all team members except one designated the Team Lead must step away from the arena.
- The judges will acknowledge that the 6 food chips nearest the park entrance are either loaded or pre-loaded.
- The remaining team member then indicates the readiness of the robot to the judge.
- When the judges give an all-clear signal, the team member should activate the robot using the single start switch and must step away from the arena immediately.

NOTE: Any interactions with the robot including but not limited to remote control, additional button presses (other than pressing the start button when directed), touching, adjusting, or hand signals to the robot, during the match could lead to disqualification. Touching the bot will end the match immediately.

- The judge will verbally indicate the start of the match to Team Lead and other Competition Volunteers helping to run the match. When this happens, the following will take place.
    - A switch will be flipped and a 3-minute countdown clock will initiate.
    - The Team lead will press the start switch or button
- The robot will have until the clock reads 2:50.00 (Two minutes and fifty seconds) to move out of the starting area. If this is not done, the judge will call the end of the run.
- The robot may complete the challenges in any order except the fireworks attraction, which, if performed, will end the round.

    - The robot may complete or skip any of the challenges prior to the 3-minute time limit.
- The match ends when one of the three scenarios occurs
    - If the remaining team member tells the judge to end the time early, the judge will stop the match and note the time remaining in the match.
    - If the robot starts the fireworks, the match will automatically end, and the clock will stop. The judge will stop the match and note the time remaining in the match.
    - If the countdown clock reaches 0 minutes and 0 seconds, the match will be ended and no remaining activities can be completed.
- After the match has ended, the team must remove their robot from playing board. If game pieces are during the match, those pieces may not be used in scoring.

7.5 Scoring

- The match will be scored as specified in Section 8.

7.6 Return to sequestration

- After the judges have scored the round, the team will quickly and safely return their robot to sequestration. After each team completes their round, each team will be able to collect their robots and make minor adjustments to compete in the next round.
- Sequestration will be re-imposed before the start of the second round of matches and the third round of matches.

VIII.Scoring

Round scores are calculated independently between matches. No points can roll over between matches. No penalties in points will be administered.

While round scores cannot have points transferred, an overall score will be the summation of the top two round scores.

After each round, a photo will be taken of the board by the judge. The judges will then be aided by the photograph. While the photo can be used to clarify, the judges will make the final call for the score for each round. The scoring distribution and board cleanup activity is given in this section.

8.1 Pre Scoring

Before any scoring occurs,

- Any game pieces that are leaning on other pieces will be removed. This does not apply to clearly built statues. Statues that are relying on another structure to remain standing will be removed. This applies to all areas of the board.
- Any game pieces that cannot be counted for points will be removed. This includes:
    - Yellow trash pieces not residing in the recycling area.
    - Blue trash pieces not residing in the recycling area.
    - Red food chips not residing in either the red enclosure, green enclosure, or recycling area.
    - Green food chips not residing in either the red enclosure, green enclosure, or recycling area.
    - Yellow ducks not residing in the pond area, recycling area, or statues.
    - The pink duck if not residing in the pond area or recycling area, or statues.

8.2 Statues

Figure 6 shows the flowchart to determine the points scored in assembling each statue. In this flow-chart, each statue score (S_{stat}) starts with an initial value of 0 points. Each statue should be standing freely, not leaning or relying on a structure other than the floor of the game board to remain upright.

NOTE: No statue can be within 2” of the park walls.

NOTE: No statue can be within 6” of another assembled statue. Single pedestals in the upright position will not count as statues.

First, the judge will determine if the statues assembled are 1, 2, or 3 pedestals tall.

- If the statues are 1 pedestal tall, 0 points will be added to the statue score.
- If the statues are 2 pedestals tall, 10 points will be added to the statue score.
    - If the 2 pedestals are in WG order (Bottom – White, Top – Green), 10 points will be added to the statue score.
    - If the 2 pedestals are in any other order, 0 points will be added to the statue score.
- If the statues are 3 pedestals tall, 20 points will be added to the statue score
    - If the 3 pedestals are in WGR order (Bottom – White, Middle – Green, Top – Red), 20 points will be added to the statue score.
    - If the 3 pedestals are in any other order, 0 points will be added to the statue score.

Second, the judge will see if a duck is placed on top of the statue. Note that the points for duck placement are not dependent on stack level points or alignment points.

- If the statue has a pink duck on top, 20 points will be added to the statue score.
- If the statue has a yellow duck on top, 10 points will be added to the statue score.
- If the statue does not have a duck on top, 0 points will be added to the statue score.

Thirdly, the judge will determine the alignment of the pedestal. There are three levels of alignment: (1) Inner – the edge of the base pedestal is within the inner circle marked on the board for statue assembly, (2) Outer – the edge of the base pedestal is within the outer circle marked on the board for statue assembly, and (3) Not Aligned – the edge of the base pedestal is outside the outer circle marked on the board for statue assembly. Figure 7 shows examples of different alignments. The points are added to the statue score as follows.

- If the statue is determined to be “Inner,” 20 points are added to the statue score.

- If the statue is determined to be “Outer,” 10 points are added to the statue score.
- If the statue is determined to be “Not Aligned”, 0 points will be added to the statue score.

After this, the judge determines the place the statue was assembled. If the statue is characterized as “Not Aligned,” the judges will skip these points listed.

- If the statue is assembled in the Duck Pond attraction area, 20 points are added to the statue score.
- If the statue is assembled in one of the Entrance statue circles, 10 points are added to the statue score.
- If the statue is assembled anywhere else in the park, 0 points are added to the statue score.

After the scores have been marked on the score sheet for one statue, the statue is disassembled by the judge and removed from the park. The judge then scores then removes each remaining statue until all statues have been scored and removed.

After all statues have been removed, all statue pedestal pieces will be removed from the board except those in the recycling areas.

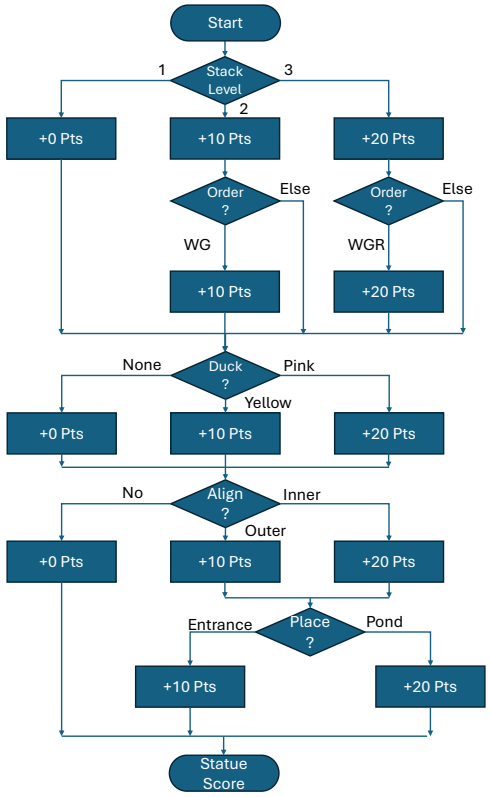


Figure 5. Statue Assembly Points Flow Chart

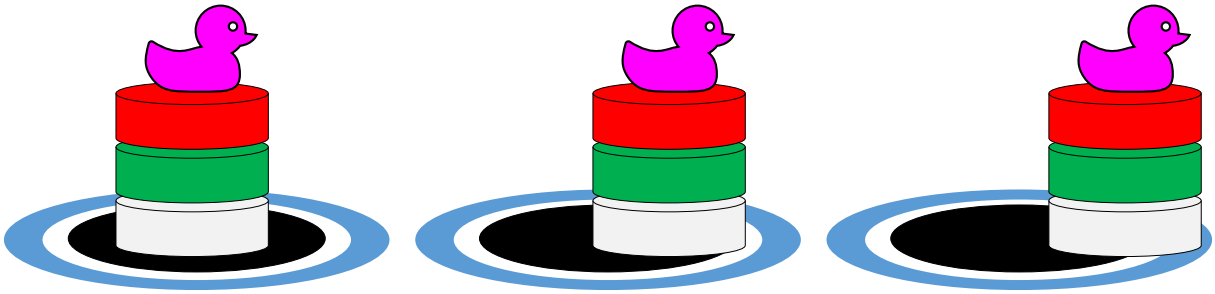


Figure 6. Alignment examples. (Left) A 3 pedestal statue classified as “Inner” aligned. (Middle) A 3 pedestal statue classified as “Outer” aligned. (Right) A 3 pedestal statue classified as “Not Aligned”

8.3 Raccoon and Alligator Enclosures

At the start of each match, the enclosure score (S_{enc}) is reset to 0.

- For each colored chip that is west in either the Raccoon or Alligator Enclosures, 5 points will be awarded to the enclosure score.

  - For each red chip west in the red enclosure (Raccoon), 11 additional points will be awarded to the enclosure score.
  - For each green chip west in the green enclosure (Alligator), 11 additional points will be awarded to the enclosure score.
- If the chips were not pre-loaded, an additional 8 points will be awarded to the enclosure score.

NOTE: If a food chip breaks the boundary of the enclosure (red to unpainted or green to unpainted), the points for that chip are not rewarded.

After the enclosure score has been marked on the score sheet, the food chips are removed by the judge from the park except for the recycling area.

8.4 Duck Pond

At the start of each match, the pond score (S_{pond}) is reset to 0. For each duck found in the pond at the end of the match, 15 points are awarded.

NOTE: If a duck breaks the inner or outer boundary (between black and blue) of the pond, the points for that duck will not be awarded. The orientation of the ducks is not a factor in awarding points. The duck can be upright, on its side, on its face, on its back, or upside down.

After the pond score has been marked on the score sheet, the ducks in the pond are removed from the park.

8.5 Recycling Area

At the start of each match, the recycling score (S_{rec}) is reset to 0.

- For each game piece (trash piece, duck, pedestal, or food chip) west in the recycling area at the end of the match, 1 point is awarded. All ducks, food chips, and pedestals are then removed from the recycling area, leaving only trash pieces in the area.

- For every trash piece remaining in the recycling area at the end of the match, 9 additional points are awarded.
- If the team has only yellow chips in the purple recycling area and only blue chips in the orange recycling area (not counting ducks, food chips, or pedestals), an additional 21 points will be awarded.

NOTE: If any item breaks the boundary of the recycling area (painted to unpainted), the points for that item will not be awarded.

After the recycling score has been marked on the score sheet, the trash pieces are removed from the entire park.

8.6 Fireworks

At the beginning of each match, the fireworks score is reset to 0 (S_{fire}). If, at the end of the match, the fireworks have been activated, 54 points will be awarded to the fireworks score.

8.7 Activity Score

At the end of each match and after the score sheets have been filled for each activity, the scores for the statues, enclosures, pond, recycling, and fireworks are added together into an activity score Act.

$$Act = S_{stat} + S_{enc} + S_{pond} + S_{rec} \tag{1}$$

NOTE: This activity score is not additional points, but a sum of all points scored from the activities of Statues, Enclosures, Duck Pond, Recycling Area, and Fireworks.

8.8 Variety Score

To encourage teams to attempt every activity within the 3-minute time limit, an additional variety score will be calculated. The variety score will be reset to 0 at the beginning of each match. The variety score can reach a maximum of 150 points and will be calculated with the following equations:

$$Var = \begin{cases} 0 & if\ V_{sum} = 0 \\ 6 & if\ V_{sum} = 1 \\ 15 & if\ V_{sum} = 2 \\ 45 & if\ V_{sum} = 3 \\ 150 & if\ V_{sum} = 4 \end{cases} \tag{2}$$

Given,

$$V_{sum} = V_{stat} + V_{enc} + V_{pond} + V_{rec}, \tag{3.a}$$

$$V_{stat} = \begin{cases} 1 & if\ S_{stat} \geq 110 \\ 0 & if\ S_{stat} < 110 \end{cases}, \tag{3.b}$$

$$V_{enc} = \begin{cases} 1 & if\ S_{enc} \geq 100 \\ 0 & if\ S_{inc} < 100 \end{cases}, \tag{3.c}$$

$$V_{pond} = \begin{cases} 1 & if\ S_{pond} \geq 80 \\ 0 & if\ S_{pond} < 80 \end{cases}, \text{ and} \tag{3.d}$$

$$V_{rec} = \begin{cases} 1 & if\ S_{red} \geq 75 \\ 0 & if\ S_{rec} < 75 \end{cases}. \tag{3.e}$$

NOTE: The fireworks are not included in the variety score due to the fireworks being a binary activity.

8.9 Time Usage Score

Students will be awarded points for adequately and effectively using their time within the 3-minute time period. The intention of this score is to reward student competitors for performing as many tasks that they can perform well in the shortest amount of time. At the end of each round, the time in seconds the students used during their round (t_{round}) will be used to calculate a time score (S_{time}). The calculation of S_{time} is calculated in equation (4).

$$S_{time} = \begin{cases} 1 & t_{round} \leq 60\ seconds \\ -\frac{1}{60}t_{round} + 2 & 60 \leq t_{round} \leq 90\ seconds \\ 0.5 & 90 \leq t_{round} \leq 150\ seconds \\ -\frac{1}{60}t_{round} + 3 & 150 \leq t_{round} \leq 180\ seconds \end{cases} \tag{4}$$

A time usage score (T_{use}) is calculated by multiplying the time score with the variety score as seen in equation (5).

$$Time = Var \times S_{time} \tag{5}$$

8.10 Overall Score

The overall score ($Total$) will be the sum of the activity score, the variety score, and the time usage score.

$$Total = Act + Var + Time$$

8.11 Tiebreakers

In the event of a tie, the teams will be ranked based on their completion time for each match. In the event multiple teams have the same points and times for a particular match, the total time for all their matches will be calculated and ranked using the same method discussed above. No additional points will be administered for fastest time. This only affects the ranking in case of ties.

8.12 Score Sheet

The score sheet is shown below in Figure 8 developed in Microsoft Excel. The yellow boxes are for data entries. This score sheet will be available on Canvas.

	Min	Sec	
Time on Screen			**OVERALL SCORE**

Statue Scores

	Level	Order	Duck	Align	Location	Score
1						
2						
3						
4						
5						
6						
7						

Statue Score

Enclosure Scores

No. Red Chips in Red Enclosure		x16 =
No. Green Chips in Green Enclosure		x16 =
No. Red Chips in Green Enclosure		x5 =
No. Green Chips in Red Enclosure		x5 =
Pre loaded chips?		Y/N

Enclosure Score

Duck Pond Scores

No. of Ducks in the Pond		x20 =

Pond Score

Recycling Area Scores

No. of Ducks, Pedestals, Food		x1 =
No. of Trash Pieces		x10 =
Separated?		

Recycling Score

Fireworks Scores

Were Fireworks activated?	

Fireworks Score

Activity Score

Variety and Time Score

Statues Attempted Score	**Time Usage Score**
Enclosure Attempted Score	
Pond Attempted Score	
Recycling Attempted Score	
Variety Score	

Figure 7. Score Sheet

IX. Communications

The hardware competition communications will be through the course Canvas site (canvas.msstate.edu). Revisions to this document will be posted there.

X. Materials and Parts

The following is a build sheet for 1 competition board.

Product	$/Unit	QTY	Total	Source	Link
Lumber ($62.30)					
RELIABILT 1-in x 4-in x 8-ft Unfinished #2 Better S4S Whitewood Common Softwood Board	8.65	3	25.95	Lowe's	(link)
Interfor 2-in x 4-in x 8-ft Euro spruce Kiln-dried Stud	3.85	1	3.85	Lowe's	(link)
Plytanium 23/32-in x 4-ft x 8-ft Rated Pine Plywood Sheathing	32.50	1	32.50	Lowe's	(link)
Screws and Nails ($15.96)					
Fas-n-Tite #6 x 2-in Yellow zinc Interior Wood screws	7.98	2	15.96	Lowe's	(link)
Sandpaper ($42.74)					
3M Multi-grade Multi-grade pack -Grit Sheet Sandpaper 9.0-in W x 11.0-in L 22 -Pack	13.88	2	27.76	Lowe's	(link)
Project Source Better 6 -Piece Polyester Paint Roller Kit	14.98	1	14.98	Lowe's	(link)
Paint ($125.86)					
HGTV Home by Sherwin-Williams (Dark Kettle Black) – 1 Quart	17.98	1	17.98	Lowe's	(link)
HGTV Home by Sherwin-Williams (Ultra White (Cab)) – 1 Quart	17.98	1	17.98	Lowe's	(link)
HGTV Home by Sherwin-Williams (Blue Shock) – 1 Quart	17.98	1	17.98	Lowe's	(link)
HGTV Home by Sherwin-Williams (Scarlet Sun) – 1 Quart	17.98	1	17.98	Lowe's	(link)
HGTV Home by Sherwin-Williams (Luscious Green) – 1 Quart	17.98	1	17.98	Lowe's	(link)
HGTV Home by Sherwin-Williams (Pumpkin Butter) – 1 Quart	17.98	1	17.98	Lowe's	(link)
HGTV Home by Sherwin-Williams (Plumb Burst) – 1 Quart	17.98	1	17.98	Lowe's	(link)
Switch ($9.67)					
Eaton 15-amp Single-pole Toggle Light Switch, White	0.98	1	0.98	Lowe's	(link)
Hubbell Hubbell RACO 8663 Drawn Handy Box, Extension, and Cover	7.86	1	7.86	Lowe's	(link)

Eaton 1 -Gang Midsize Size White Polycarbonate Indoor Toggle Wall Plate	0.83	1	0.83	Lowe's	(link)
Prefabricated Game Pieces ($39.97)					
Chivao 12 Pc Rubber Ducks with 12 Sunglasses (Yellow)	9.99	1	9.99	Amazon	(link)
Chivao 12 Pc Rubber Ducks with 12 Sunglasses (Pink)	8.99	1	8.99	Amazon	(link)
500 pieces plastic poker chips, 37 mm 10 colors (500 pieces)	17.99	1	17.99	Amazon	(link)
3D Printer Materials ($70.99)					
OVERTURE TPU 95A Filament 1.75mm, Flexible 95A Soft 3D Printer Filament, 1kg Spool (2.2 lbs), Dimensional Accuracy +/- 0.02 mm, Fit Most FDM Printers (TPU White)	23.79	1	23.79	Amazon	(link)
TPU Filament 1.75mm, Green Flexible TPU 3D Printer Filament with 20g Cleaning Filament, 2.0lbs Spool, Dimensional Accuracy +/- 0.05mm, 95A Soft TPU Green	23.99	1	23.99	Amazon	(link)
YOYI YOYI 3D Printer Filament, 95A TPU Filament 1.75mm, Flexible Filament, 0.8KG Spool,Dimensional Accuracy +/- 0.03 mm, Soft TPU Filaments of High Elasticity (Red)	22.99	1	22.99	Amazon	(link)
Electronics ($161.56)					
Raspberry Pi 4 Model B 2019 Quad Core 64 Bit WiFi Bluetooth (4GB)	62.99	1	62.99	Amazon	(link)
Argon USB C Power Supply with Switch | Raspberry Pi 4 Power Supply	12.00	1	12.00	Amazon	(link)
Mini HDMI to HDMI Cable (5')	6.59	1	6.59	Amazon	(link)
Micro SD Cards (10 Pack) 64 GB	39.99	2	79.98	Amazon	(link)
Total Per Board					
		Total	**$529.05**		

The Capstone instructors and competition organizers are asking for two boards to be built. Because some of these devices are not needed to be purchased again in developing a second board, the following has been west out of the estimate for the second board:

- Paint ($125.86)
- Pink Ducks ($8.99)
- 3D printing filament ($70.99)
- Electronics ($161.56)

To make it easier to reset the competition board on competition day, a second switch is requested to allow the judge to quickly reset the time clock and fireworks video. The following has been added to the cost of a second board:

- Switch ($9.67)

The estimated cost of building one board for capstone student practice and one board for capstone competition is $700.37.

XI. Schedule

This is a tentative schedule. This schedule is subject to change and any changes will be logged into the Change Log.

Date	Milestone
8/20/2025	• Classes Begin. • FAQ questionnaire opened for Revision 1.
9/15/2025	• FAQ for Revision 1 closed.
10/1/2025	• Release of Revision 1 based on FAQ questionnaire. • FAQ questionnaire re-opened for Revision 2 (Final Revision)
11/21/2025	• FAQ questionnaire closed for Revision 2
12/1/2025	• Capstone I Subsystem Demonstration
1/14/2026	• Classes Resume.
1/21/2025	• Release of Revision 2 based on FAQ questionnaire.
(TBD)/2026	• Competition Day

XII. Revisions

Revision	Date	Notes
0.0	5/2/2025	• Initial Release to Capstone TA and Co-instructor
0.1	5/7/2025	• Scoring updated to make max score for each category to be 100 (except Fireworks). These changes are reflected in the narrative, the flowchart figure, and the score sheet figure. Also updated variety score to be 100 points and updated the method to calculate the variety score. • Formatting updates were made to take care of a few careless errors.
0.2	6/6/2025	• Fixed some grammar and spelling mistakes from Amy Barton's and Chase Robinson's Review

		• New figure 1 of playing field. Contains a board with randomized items with no measurements and one without randomized items but with measurements.
		• Updated document to have new ducks
		• Updated the color of trash pieces to be yellow and blue (anticipation of new board design)
		• Updated description of statue locations
		• Updated cardinal directions of (Top, Bottom, Left, Right) to (North, South, West, and East)
		• Updated document to use different chips
		• Removed Firework video submissions by students
		• Reorganized VI section into “Must”, “Should”, “May”, and “Cannot”
		• Expanded and reorganized VII to include sequestration, pre match, match, scoring statement, and return to sequestration.
		• Reorganized and rewrote VIII scoring section
		  ○ New weights of points for each activity
		  ○ Time usage score
		  ○ Pre-scoring section
		• New Score Sheet
		• Updated prices and materials in building list
		• Added estimate for 2 boards
		• Schedule adjusted

XIII.FAQ

As the FAQ questionnaire is filled out and as subsequent revisions are released, this section will be updated.